Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A reaction mixture containing $\mathrm{H_2}$, $\mathrm{N_2}$ and $\mathrm{NH_3}$ has partial pressures $2\ \mathrm{atm}, 1\ \mathrm{atm}$ and $3\ \mathrm{atm}$ respectively at $725\ \mathrm{K}$. If the value of $\mathrm{K_p}$ for the reaction, $\mathrm{N_{2(\ g)} + 3H_{2(\ g)} \rightleftharpoons 2NH_{3(\ g)}}$ is $4.28 \times 10^{-5}\ \mathrm{atm^{-2}}$ at $725\ \mathrm{K}$, in which direction the net reaction will go ?

(A) Forward

(B) Backward

(C) No net reaction

(D) Direction of reaction cannot be predicted

Q.2

$\mathrm{K_1}$ and $\mathrm{K_2}$ are equilibrium constant for reactions (1) and (2)
$\mathrm{N_2(\ g) + O_2(\ g) \rightleftharpoons 2NO(g)}$ ... (1)
$\mathrm{NO(g) \rightleftharpoons \frac{1}{2}\ N_2(\ g) + \frac{1}{2}O_2(\ g)}$ ... (2)
Then,

(A) $\mathrm{K_1 = \left(\frac{1}{K_2}\right)^2}$

(B) $\mathrm{K_1 = K_2^2}$

(C) $\mathrm{K_1 = \frac{1}{K_2}}$

(D) $\mathrm{K_1 = (K_2)^0}$

Q.3

For the reaction

$$\mathrm{2HI(g) \rightleftharpoons H_2(\ g) + I_2(\ g)}$$

The degree of dissociation (α) of $\mathrm{HI(g)}$ is related to equilibrium constant, K_p by the expression

(A) $\mathrm{\frac{1+2\sqrt{K_p}}{2}}$

(B) $\mathrm{\sqrt{\frac{1+2\ K_p}{2}}}$

(C) $\mathrm{\sqrt{\frac{2\ K_p}{1+2\ K_p}}}$

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(D) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

Q.4

Given are the information:
(i) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g); \quad K_{p1}$
(ii) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad K_{p2}$
(iii) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g); \quad K_{p3}$
The expression of K_p for the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ is

(A) $K_{p2}^{1/2} K_{p3} / K_{p1}^{3/2}$

(B) $K_{p1} K_{p2}^{1/2} / K_{p3}^{3/2}$

(C) $K_{p2} K_{p3}^{1/2} / K_{p1}^{3/2}$

(D) $K_{p2} K_{p3}^{1/2} / K_{p1}^{1/2}$

Q.5

At temperature T, a compound $AB_2(g)$ dissociates according to the reaction $2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$ with degree of dissociation α, which is small compared with unity. The expression for K_p, in terms of α and the total pressure, P_T is

(A) $\frac{P_T \alpha^3}{2}$

(B) $\frac{P_T \alpha^2}{3}$

(C) $\frac{P_T \alpha^3}{3}$

(D) $\frac{P_T \alpha^2}{2}$

Q.6

For the reaction:

$$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

The equilibrium constant $K_p = 2.9 \times 10^{-5}\ atm^3$. The total pressure of gases at equilibrium when 1.0 mole of reactant was heated will be

(A) $0.0194\ atm$

(B) $0.0388\ atm$

(C) $0.058\ atm$

(D) $0.0667\ atm$

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Q.7

Reaction between iron and steam is reversible if it is carried out

$$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$$

(A) at constant T

(B) at constant P

(C) in an open vessel

(D) in a closed vessel

Q.8

In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be

(A) Reduced to half of its original value

(B) becomes (original)/4

(C) doubled

(D) constant

Q.9

For which of the following K_p may be equal to 0.5 atm

(A) $2HI \rightleftharpoons H_2 + I_2$

(B) $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

(C) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

(D) $2NO_2 \rightleftharpoons N_2O_4$

Q.10

If pressure is applied to the following equilibrium, liquid $\rightleftharpoons$ vapours the boiling point of liquid

(A) will increase

(B) will decrease

(C) may increase or decrease

(D) will not change

Q.11

If $CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(1)}$

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$K_p = 1.086 \times 10^{-4}\ atm^2$ at $25^\circ C$. The efflorescent nature of $CuSO_4.5H_2O$ can be noticed when vapour pressure of H_2O in atmosphere is

(A) > 7.29 mm

(B) < 7.92 mm

(C) ≥ 7.92 mm

(D) None

Q.12

The enthalpies of two reaction are ΔH_1 and ΔH_2 (both positive) with $\Delta H_2 > \Delta H_1$. If the temperature of the reacting system is increased from T_1 to T_2, predict which of the following alternatives is correct?

(A) $\frac{K_1'}{K_1} = \frac{K_2'}{K_2}$

(B) $\frac{K_1'}{K_1} > \frac{K_2'}{K_2}$

(C) $\frac{K_I'}{K_I} < \frac{K_2'}{K_2}$

(D) None

Q.13

The reaction: $3O_2 \rightleftharpoons 2O_3$ $\Delta H = +69,000$ calories is favoured by:

(A) high temperature and low pressure

(B) high temperature and high pressure

(C) low temperature and high pressure

(D) low temperature and low pressure

Q.14

The equilibrium constant at $323^\circ C$ is 1000 . What would be its value in the presence of a catalyst in the forward reaction?

$$A + B \rightleftharpoons C + D + 38Kcal$$

(A) $1000\times$ concentration of catalyst

(B) 1000

(C) $\frac{1000}{\text{concentration of catalyst}}$

(D) impossible to tell

Q.15

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Which of the following reactions represent a heterogeneous equilibrium?

(A) $(a)H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(B) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

(C) $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

(D) $2O_{3(g)} \rightleftharpoons 3O_{2(g)}$

Q.16

For a reaction if $K_p > K_c$, the forward reaction is favoured by

(A) Low pressure

(B) High pressure

(C) High temperature

(D) Low temperature

Q.17

Naphthalene, a white solid used to make mothballs, has a vapour pressure of $0.10 mmHg$ at $27^\circ C$. Hence, and K_p and K_c for the equilibrium are :

$$C_{10}H_8(s) \rightleftharpoons C_{10}H_8(g)$$

(A) $0.10, 0.10$

(B) $0.10, 4.1 \times 10^{-3}$

(C) $1.32 \times 10^{-4}, 5.34 \times 10^{-6}$

(D) $5.36 \times 10^{-6}, 1.32 \times 10^{-4}$

Q.18

According to law of mass action rate of a chemical reaction is proportional to

(A) Concentration of reactants

(B) Molar concentration of reactants

(C) Concentration of products

(D) Molar concentration of products

Q.19

When two reactants, A and B are mixed to give products C and D, the reaction quotient, Q at the final stages of the reaction

(A) is zero

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(B) decreases with time

(C) is independent of time

(D) increases with time

Q.20

Which of the following will not change the concentration of ammonia at the equilibrium?

(A) Increase of pressure

(B) Increase of volume

(C) Addition of catalyst

(D) Decrease of temperature

Q.21

An equilibrium system for the reaction between hydrogen and iodine to give hydrogen iodide at $765\ \mathrm{K}$ in a 5 litre volume contains 0.4 mole of hydrogen, 0.4 mole of iodine and 2.4 moles of hydrogen iodide. The equilibrium constant for the reaction is: $\mathrm{H}_{2(\ g)}+\mathrm{I}_{2(\ g)} \rightleftharpoons 2\mathrm{HI}_{(g)}$, is

Q.22

$2.0\ \mathrm{mol}$ of PCl_5 were introduced in a vessel of $5.0\ \mathrm{L}$ capacity at a particular temperature. At equilibrium, PCl_5 was found to be 35% dissociated into PCl_3 and Cl_2. The value of $\mathrm{K}_c \times 100$ for the reaction is

Q.23

In equilibrium reaction, x moles of the reaction A decompose to give 1 mole each of C and D. If the fraction of A decomposed at equilibrium is independent of initial concentration, then the value of x would be

Q.24

At a certain temperature the following equilibrium is established,

$$\mathrm{CO(g)}+\mathrm{NO}_2(\ \mathrm{g}) \rightleftharpoons \mathrm{CO}_2(\ \mathrm{g})+\mathrm{NO(g)}$$

One mole of each of the four gases is mixed in one litre container and the reaction is allowed to reach equilibrium state. When excess of baryta water is added to the equilibrium mixture, the weight of white precipitate obtained is 236.4 g. The equilibrium constant, K_C of the reaction is p, then the value of $4p$ is

Q.25

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For the reaction

$$\mathrm{PCl}_{3(g)} + \mathrm{Cl}_{2(g)} \rightleftharpoons PCl_{5(g)}$$

The value of K_c at $250^{\circ}\mathrm{C}$ is 26 . The value of $\mathrm{K_p} \times 100$ at this temperature will be

Q.26

The equilibrium constant for the reaction, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO$ is $4 * 10^{-4}$ at $2000K$
In presence of a catalyst, equilibrium is attained ten times faster. Therefore, the equilibrium constant, in presence of the catalyst at $2000\ \mathrm{K}$ is $\mathrm{k} \times 4 \times 10^{-4}$, then the value of k is

Q.27

The vapour density of undecomposed $\mathrm{N_2O_4}$ is 46. When heated, vapour density decreases to 24.5 due to its dissociation to $\mathrm{NO_2}$. The $\%$ dissociation of $\mathrm{N_2O_4}$ at the final temperature is

Q.28

At a certain temperature 2 moles of carbon monoxide and 3 moles of chlorine were allowed to reach equilibrium according to the reaction $\mathrm{CO} + \mathrm{Cl_2} \rightleftharpoons \mathrm{COCl_2}$ in a 5 lit vessel. At equilibrium if one mole of CO is present then equilibrium constant for the reaction is x then the value of $10\mathrm{x}$ is:

Q.29

The initial pressure of $\mathrm{COCl_2}$ is 1000 torr. The total pressure of the system becomes 1500 torr, when the equilibrium $\mathrm{COCl_2(\ g) \rightleftharpoons CO(g) + Cl_2(\ g)}$ is attained at constant temperature. The value of K_p of the reaction is

Q.30

For the equilibrium

$$\mathrm{NH_2COONH_4(\ s) \rightleftharpoons 2NH_3(\ g) + CO_2(\ g)}$$

$\mathrm{Pco_2} = 1\ \mathrm{atm}$ at $100^{\circ}\mathrm{C}$. Hence, equilibrium constant is (in atm ^{3}):

Answers & Solutions

Q.1 Answer:

Backward

Solution:

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$$Q_p = \frac{(P_{NH_3})^2}{P_{N_2} \times (P_{H_2})^3} = \frac{(3)^2}{(1)(2)^3} = \frac{9}{8}\ atm^{-2}$$

$$= 1.125\ atm^{-2}$$

Since value of Q_p is larger than K_p $\left(4.28 \times 10^{-5}\ atm^{-2}\right)$, it indicates net reaction will proceed in backward direction.

Q.2 Answer:

$K_1 = \left(\frac{1}{K_2}\right)^2$

Solution:

$N_2 + O_2 = 2NO \quad \ldots K_1$

$2NO \rightleftharpoons N_2 + O_2 \quad \ldots \quad \frac{1}{K_1}$

$NO \rightleftharpoons \frac{1}{2} N_2 + O_2 \quad \ldots K_2 = \left(\frac{1}{K_1}\right)^{1/2}$

$K_2 = \left(\frac{1}{K_1}\right)^{1/2} \quad \text{or} \quad K_2^2 = \frac{1}{K_1}$

$K_1 = \frac{1}{(K_2)^2}$

Q.3 Answer:

$\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

Solution:

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

$1-\alpha$	$\frac{\alpha}{2}$	$\frac{\alpha}{2}$

$K_p = \frac{\left(\frac{\alpha}{2} P_T\right)^2}{(1-\alpha)^2 P_T^2}$

$\frac{\alpha}{1-\alpha} = 2\sqrt{K_p}$

$\alpha = \frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

Q.4 Answer:

$K_{p2} K_{p3}^{1/2} / K_{p1}^{3/2}$

Solution:

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Given equation is obtained by the manipulations Eq.(ii) $-\frac{3}{2}$ Eq. (i) $+\frac{1}{2}$ Eq. (iii)

Hence, $K_p = \frac{K_{p_2} K_{p3}^{1/2}}{K_{p1}^{3/2}}$

Q.5 Answer:

$\frac{P_T \alpha^3}{2}$

Solution:

For the given equilibrium, the equilibrium concentration are

$$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$$

Equilib. conc.	$c(1-\alpha)$	$c\alpha$	$\frac{c\alpha}{2}$

$$K_P = \frac{(P_{B_2})(P_{AB})^2}{(P_{AB_2})^2} = \frac{\frac{c\alpha}{2} \times (c\alpha)^2 \times P_T}{[c(1-\alpha)]^2 \left[c\left(1+\frac{\alpha}{2}\right)\right]}$$

$$K_P = \frac{\alpha^3 \times P_T}{2(1-\alpha)^2\left(1+\frac{\alpha}{2}\right)}$$

Since, α is small compared to unity,

so $1-\alpha \simeq 1$ and $1+\frac{\alpha}{2} \simeq 1$

$$K_P = \frac{\alpha^3 \times P_T}{2}$$

Q.6 Answer:

0.058 atm

Solution:

$$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$$

Moles initially	1	0	0
Moles at equib.	$(1-x)$	$2x$	x

Total moles of gaseous substances at equilibrium $= 3x$

Equilibrium pressure $= P_T$

$P_{NH_3} = \frac{2}{3}P_T, P_{CO_2} = \frac{P_T}{3}$

$K_P = (P_{NH_3})^2 P_{CO_2}$

$2.9 \times 10^{-5} = \frac{4(P_T)^3}{27}$

$P_T = 0.058$ atm

Q.7 Answer:

in a closed vessel

Solution:

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In an open vessel H_2 gas will escape. So the reaction will be reversible if it is carried out in a closed vessel.

Q.8 Answer:

constant

Solution:

K_c or K_p do not depend on concentration, but only on temperature

Q.9 Answer:

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Solution:

For $K_p = 0.5$ **atm**

$\Delta n = 1$ **(since the unit is atm)**

and $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$\Delta n = 1$

Q.10 Answer:

will increase

Solution:

Boiling point of a liquid is the temperature at which vapour pressure became equal to atm pressure. If the pressure is applied to the above equilibrium the reaction will go to the backward direction, i.e. vapour pressure decreases hence the boiling point increases

Q.11 Answer:

< 7.92 mm

Solution:

An efflorescent salt is one that loses H_2O to the atmosphere.

For the reaction

$CuSO_4 \cdot 5H_2O_{(s)} + CuSO_4 \cdot 3H_2O_{(s)} + 2H_2O_{(i)}$

$K_p = \left(p'_{H_2O}\right)^2 = 1.086 \times 10^{-4}$

$p'_{H_2O} = 1.042 \times 10^{-2}$ atm $= 7.92$ mm

Q **If** p'_{H_2O} **at** $25^\circ C < 7.92$ mm **only then, reaction will proceed in forward direction,**

Q.12 Answer:

$\frac{K'_I}{K_I} < \frac{K'_2}{K_2}$

Solution:

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As the temperature of reacting system is increased the equilibrium constant of reaction is also increased for endothermic reactions so for two reactions on increasing the temperature by equal amounts $\frac{\mathrm{K}_1'}{\mathrm{K}_1} < \frac{\mathrm{K}_2'}{\mathrm{K}_2}$

Q.13 Answer:

high temperature and high pressure

Solution:

According to Le Chatalier principle, formation of ozone is favoured by high temperature (endothermic reaction) and high pressure.

Q.14 Answer:

1000

Solution:

The equilibrium constant varies only with temperature. At constant temperature it will not vary.

Q.15 Answer:

$\mathrm{CaCO}_{3(s)} \rightleftharpoons \mathrm{CaO}_{(s)} + \mathrm{CO}_{2(g)}$

Solution:

In heterogeneous equilibrium, the physical state of all the reactants and products are not the same.

Q.16 Answer:

Low pressure

Solution:

For a reaction if $K_p > K_c$, the forward reaction is favoured by low pressure.

$$K_p = K_c(RT)^{\Delta n}$$

Since $K_p > K_c$

$\therefore \Delta \mathrm{n} > 0$, i.e. product side contains more particles than the reactant side.

At low pressure, the equilibrium will shift in the forward direction so that more and more gaseous products are formed. This will increase the total pressure and nullify the effect of low pressure.

Q.17 Answer:

$1.32 \times 10^{-4}, 5.34 \times 10^{-6}$

Solution:

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$K_p = P_{C_{10}H_8(g)}$
$= 0.10\ mm = \frac{0.10}{760}\ atm = 1.32 \times 10^{-4}$
$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{K_p}{RT} = \frac{1.32 \times 10^{-4}}{0.0821 \times 300}$
$= 5.36 \times 10^{-6}$

Q.18 Answer:

Molar concentration of reactants

Solution:

According to law of mass-action, "at a given temperature, the rate of a reaction at a particular instant is proportional to the product of the active masses of the reactants at that instant raised to powers which are numerically equal to the numbers of their respective molecules in the stoichiometric equation describing the reaction".

Q.19 Answer:

is independent of time

Solution:

In the final stage of the reaction, the reaction is at equilibrium and thus the value of Q remains constant with time.

Q.20 Answer:

Addition of catalyst

Solution:

Catalyst does not disturb the equilibrium state.

Q.21 Answer:

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Solution:

The concentration of various species at equilibrium are :
$[H_2] = \frac{0.4}{5} = 0.08\ mol\ L^{-1}$
$[I_2] = \frac{0.4}{5} = 0.08\ mol\ L^{-1}$
$[HI] = \frac{2.4}{5} = 0.48\ mol\ L^{-1}$
$K = \frac{[HI]^2}{[H_2][I_2]}$
Substituting the various values in the above expression of K, we get
$K = \frac{(0.48)^2}{(0.08)(0.08)} = 36.0$

Q.22 Answer:

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Solution:

Moles of PCl_5 dissociated $= \frac{2\times35}{100} = 0.7$\n**Moles of PCl_5 left undissociated** $= 2 - 0.7$\n$=$ 1.3 mol\n$[PCl_5] = \frac{1.3}{5}M$, $[PCl_3] = \frac{0.7}{5}M$, $[Cl_2]$\n$= \frac{0.7}{5}M$\n**K** $= \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{0.7}{5}\right)\left(\frac{0.7}{5}\right)}{\left(\frac{1.3}{5}\right)}$

$k_c = .075$

$k_c \times 100 = 7.5$

Q.23 Answer:

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Solution:

The fraction of A decomposed at equilibrium is independent of initial concentration, meaning the equilibrium constant expression is free from concentration term. The equilibrium reaction is

	xA	$\rightleftharpoons$	C	+	D
Initial conc.	c		0		0
Conc. at. equil.	$c(1-\alpha)$		$\underset{x}{c\alpha}$		$\underset{x}{c\alpha}$

$K_C = \frac{[C][D]}{[A]^x} = \frac{\left(\frac{c\alpha}{x}\right)\times\left(\frac{c\alpha}{x}\right)}{[c(1-\alpha)]^x} = \frac{(c\alpha)^2}{x^2[c(1-\alpha)]^x}$

The expression of K_C would be free from concentration term only when value of x is 2 . Putting $x = 2$, gives

$K_C = \frac{c^2\alpha^2}{4[c(1-\alpha)]^2} = \frac{\alpha^2}{4(1-\alpha)^2}$

Q.24 Answer:

—

Solution:

$$CO(g) + NO_2(g) \rightleftharpoons CO_2(g) + NO(g).$$

Moles initially	1	1	1	1
Moles at equib.	$1-x$	$1-x$	$1+x$	$1+x$

$CO_2 + Ba(OH)_2 \longrightarrow BaCO_3 + H_2O$

Moles of $BaCO_3$ $= \frac{236.4}{197} = 1.2$

Moles of CO_2 at equilibrium $= 1.2$ **or,** $1 + x = 1.2; x = 0.2$

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$\therefore \quad K_C = \left(\frac{1+x}{1-x}\right)^2 = \left(\frac{1.2}{0.8}\right)^2 = 2.25$
$p = 2.25, 4p = 9$

Q.25 Answer:

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Solution:

$K_p = K_c(RT)^{\Delta n} = 26 \times (0.082 \times 523)^{-1} = 0.61$

Q.26 Answer:

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Solution:

Equilibrium is constant at constant temperature for a reaction
Therefore the equilibrium constant in the presence of a catalyst at $2000\ K$ is the same as the equilibrium constant in the absence of a catalyst at $2000\ K$. It is 4×10^{-4}
The presence of a catalyst does not affect the value of the equilibrium constant. It only increases the rates of forward and reverse reaction to equal extent.

Q.27 Answer:

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Solution:

We know that,
Vapour density of N_2O_4 is 46 (decomposed)
Vapour density of N_2O_4 is 24.5 (after having N_2O_4)
We know that the molecular weight M_w can be expressed in terms of vapour density,
$M_w = 2\times$ **Vapour Density**
Now we have to find the molecular weight before the dissociation. So, the molecular weight before the dissociation is,
$M_w = 2\times$ **Vapour Density** $= 2 \times 46 = 92$
So, now the molecular weight after the dissociation will be
$M_w = 2\times$ **Vapour Density** $= 2 \times 24.5 = 49$
Now we have to find the Van't Hoff factor, which can also be denoted as i. So, the Van't Hoff faction is,
Van't Hoff's factor $= \frac{\text{Observed value}}{\text{Actual value}} = \frac{92}{49} = 1.877 \simeq 1.88$
We know that the degree of dissociation α is given in terms of Van't Hoff's factor 'i' as follow:
$\alpha = \frac{i-1}{n-1}$
In the above-mentioned formula,
α is the degree of dissociation, n is the number of particles after the dissociation.
So, we are given that nitrogen trioxide dissociates into nitrogen dioxide. The reaction is as shown below:

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$N_2O_4 \rightarrow 2NO_2$

In the above-mentioned reaction, the value of n is 2 .

So, putting the values in the formula of degree of dissociation.

We know that, $i = 1.88$ and $n = 2$. So, the dissociation constant is obtained as follows:

$\alpha = \frac{i-1}{n-1} = \frac{1.88-1}{2-1} = 0.88$

There the percentage dissociation of N_2O_4 into the NO_2 equals to the,

$\alpha^0\% = 0.88 \times 100^\circ\% = 88^\circ\%$

So the closest answer is 88%.

So, the correct answer is Option D.

Note: When particles associates in the solution, the values of 'i' is:

i $\langle 1$, 'i' is less than unity.

Example, benzoic acid in benzene forms dimer

When the solute dissociates in the solution, then the value of 'i' is:

i $\rangle 1$, 'i' is greater than the unity.

For example, sodium chloride is a solution to form sodium and chloride ions.

When solute does not dissociate or associate in the solution, the value of 'i' is:

$i = 1$, 'i' is equal to the unity ,For example, Glucose in water.

Q.28 Answer:

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Solution:

	CO +	Cl_2 $\longrightarrow$	$COCl_2$
At t = 0	2	3	0
At equilibrium	(2–1)	(3–1)	1
Concentrations (V = 5 lit)	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$

$K_c = \frac{[COCl_2]}{[CO][Cl_2]} = \frac{1/5}{\frac{1}{5}\times\frac{2}{5}} = \frac{25}{10}$ **or** $K_c = 2.5$

Q.29 Answer:

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Solution:

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$$\underset{(1000-p)}{COCl_2(g)} \rightleftharpoons \underset{p}{CO(g)} + \underset{p}{Cl_2(g)}$$

.....At equilibrium.

Total pressure at equilibrium

$= (1000 - p) + p + p = 1000 + p = 1500$

$\therefore \quad p = 500$ **torr**

$K_p = \frac{pCO \times pCl_2}{pCOCl_2} = \frac{500 \times 500}{500} = 500$

Q.30 Answer:

—

Solution:

Only gaseous reactants are considered.

$$\underset{}{NH_2COONH_4(s)} \rightleftharpoons \underset{2x}{2NH_3(g)} + \underset{x}{CO_2(g)}$$

$p_{CO_2} = \frac{p_{Total}}{3} = 1 \text{ atm}$

$p_{NH_3} = \frac{2p_{Total}}{3} = 2 \text{ atm}$

$K_p = p_{NH_3}^2 \cdot p_{CO_2} = 4 \text{ atm}$

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